Why transient

Transient analysis steps time: capacitors use companion models at each dt

The idea

- GOLDEN: dt=0.5, 4 steps, final_v2≈2.255181 (BE RC from v2_init=0)

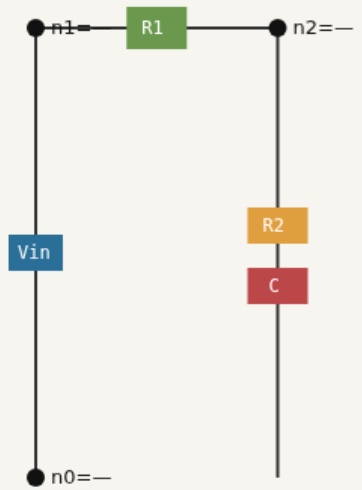
Companion

TRANSIENT BE

Companion

Step 1 / 5 · comp · module04-01-transient-be

elements=5



Explanation

C → G=C/dt + history current.

- BE

```
C/dt
```

Companion

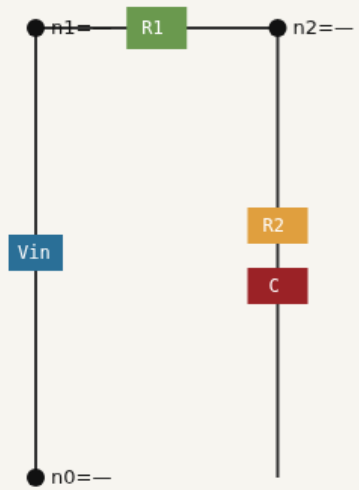
dt

TRANSIENT BE

dt

Step 2 / 5 · dt · module04-01-transient-be

dt=0.5



Explanation

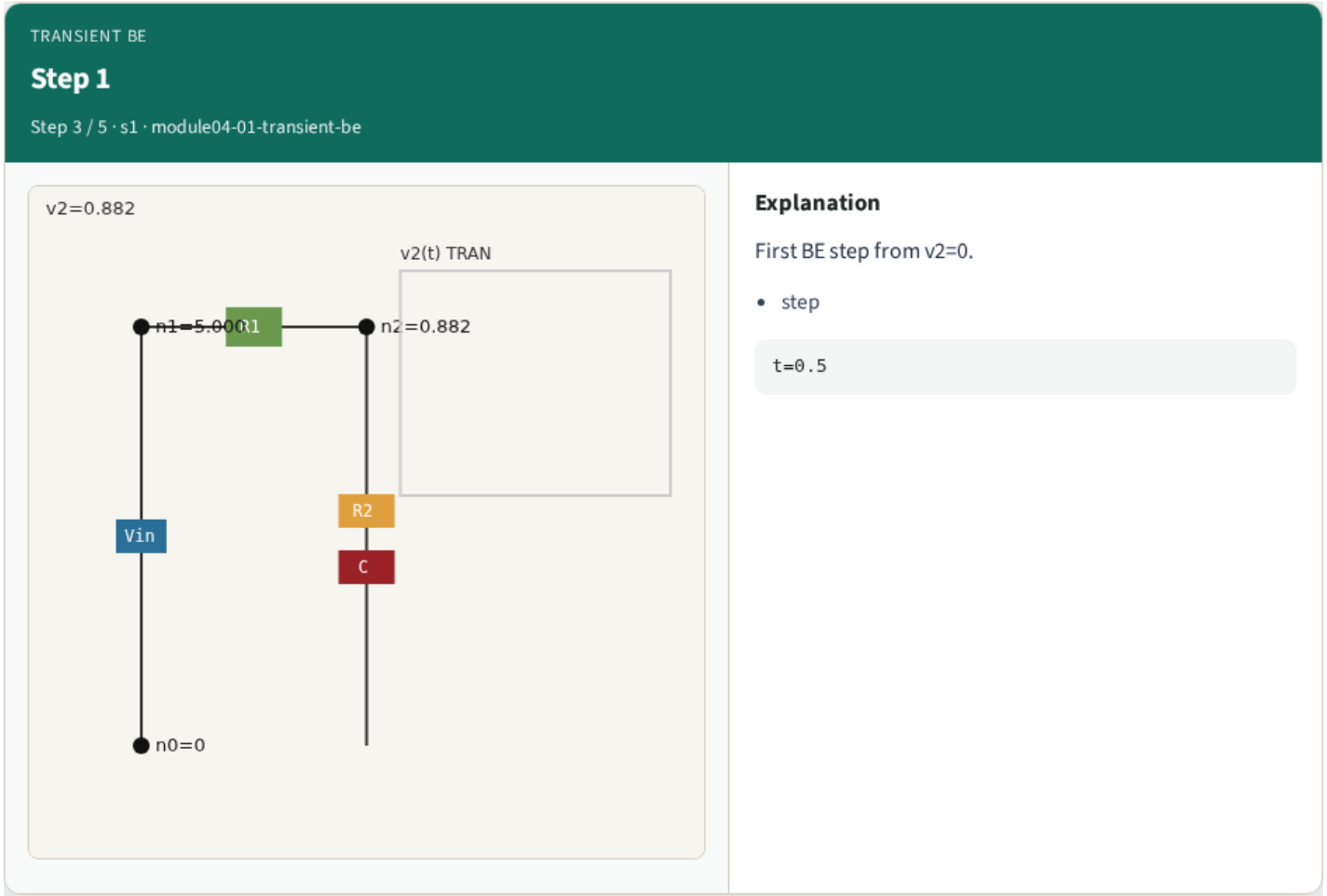
dt=0.5 on the toy.

- dt

```
0.5
```

dt

Step 1


TRANSIENT BE
Step 1
Step 3 / 5 · s1 · module04-01-transient-be
v2=0.882
v2(t) TRAN
n1=5.000
R1
n2=0.882
R2
Vin
C
n0=0
Explanation
First BE step from v2=0.
step
t=0.5

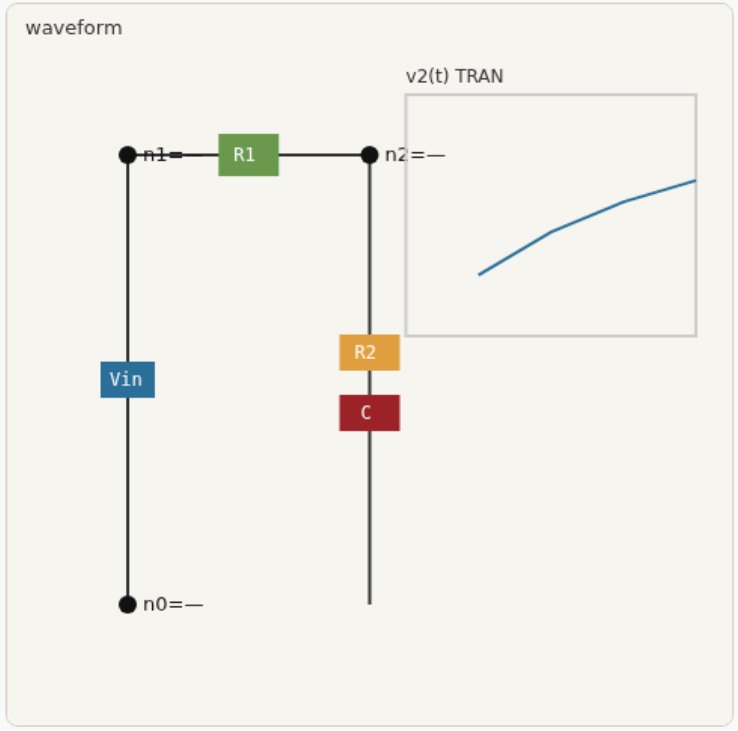
Step 1

Waveform

TRANSIENT BE

Waveform

Step 4 / 5 · wav · module04-01-transient-be



Explanation

Four steps toward DC.

- wave

```
4
```

Waveform

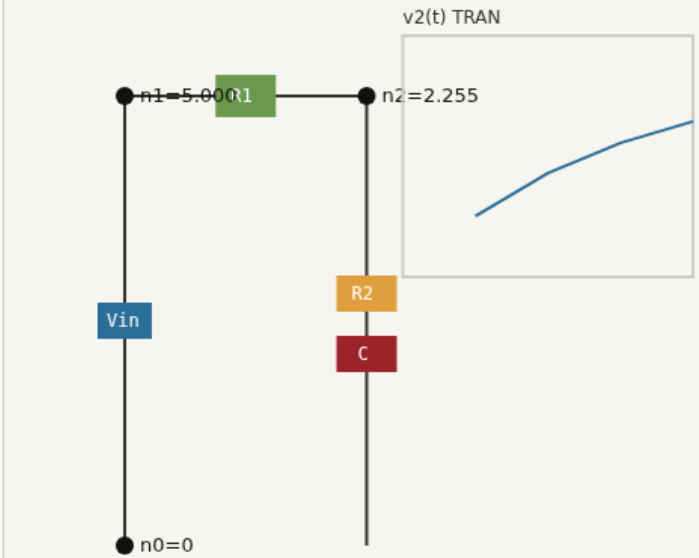
Final

TRANSIENT BE

Final

Step 5 / 5 · fin · module04-01-transient-be

final=2.255181



Explanation

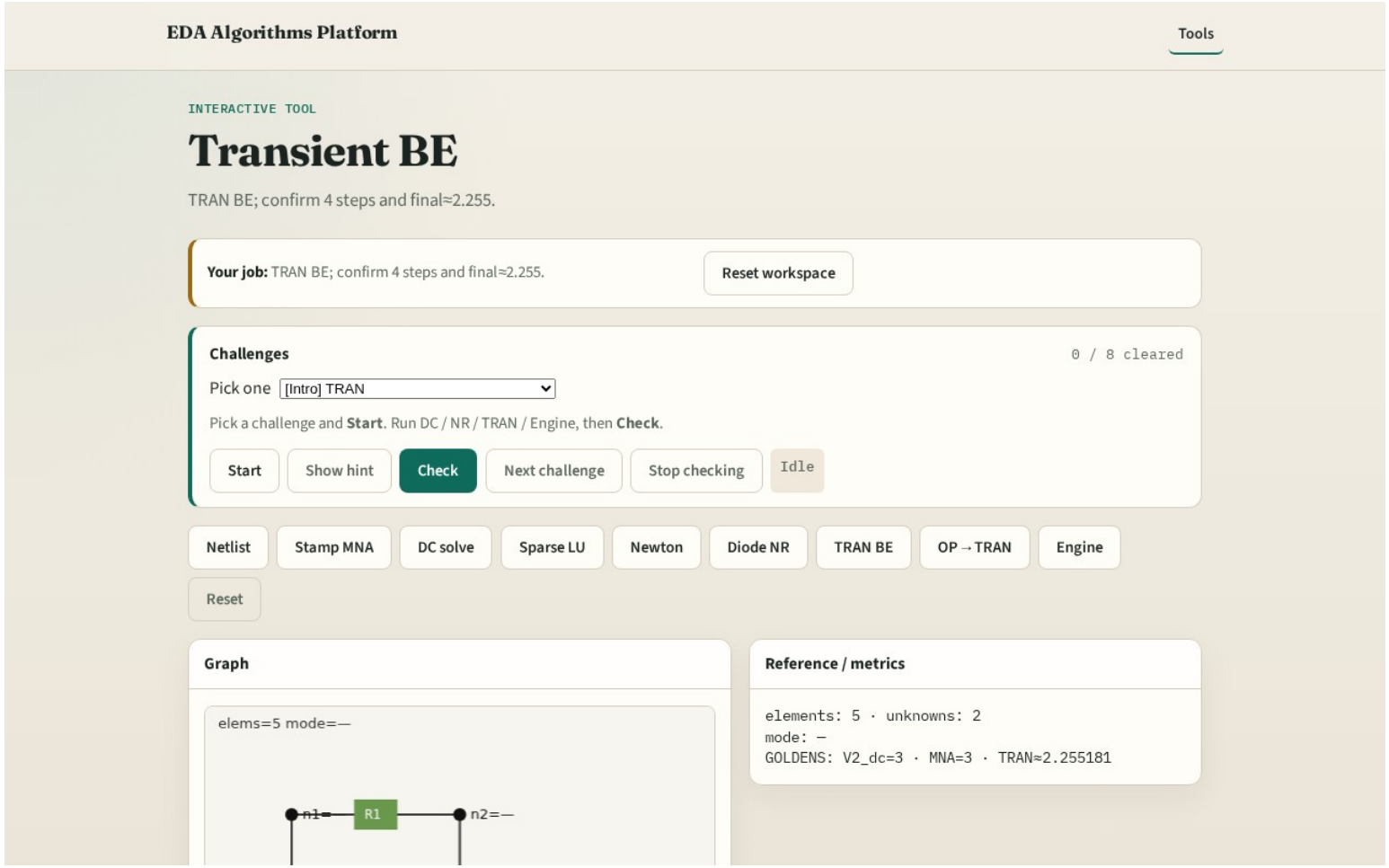
GOLDEN final≈2.255.

- GOLDEN

```
2.255
```

Final

Browser lab track



Browser lab starter

Implement track

- Call transient_be(dt=0.5, steps=4); assert abs(final_v2-2.255181)<1e-5

Pitfalls

- Using forward Euler by mistake
- Wrong initial capacitor voltage

Your turn

- Next: OP then TRAN pipeline